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Assignment 21: Data Loading in ETL

Dataset:

Order_ID	Customer_ID	Sales_Amount	Order_Date
O101	C001	4500	12-01-2024
O102	C002	Null	15-01-2024
O103	C003	3200	2024/01/18
O101	C001	4500	12-01-2024
O104	C004	Three Thousand	20-01-2024
O105	C005	5100	25-01-2024

Q1. Data Understanding

Identify all data quality issues present in the dataset that can cause problems during data loading.

Answer:

The dataset contains the following data quality issues:

1. Duplicate Primary Key (O101 appears twice).

2. Missing Value in Sales_Amount (O102).

3. Invalid Data Type ("Three Thousand" is text instead of numeric).

4. Inconsistent Date Formats (DD-MM-YYYY and YYYY/MM/DD).

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Q2. Primary Key Validation

Assume Order_ID is the Primary Key.

a) Is the dataset violating the Primary Key rule?

b) Which record(s) cause this violation?

Answer:

Assuming Order_ID is the Primary Key:

1. Yes, the dataset violates the Primary Key rule because O101 appears twice.
2. The duplicate records with Order_ID O101 cause the violation.

Q3. Missing Value Analysis

Which column(s) contain missing values?

a) List the affected records

b) Explain why loading these records without handling missing values is risky

Answer:

The Sales_Amount column contains missing values.

Affected Record: Order_ID O102.

Risk: Incorrect total sales calculation and possible database rejection if NOT NULL constraint exists.

Q4. Data Type Validation

Identify records where Sales_Amount violates expected data type rules.

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a) Which record(s) will fail numeric validation?

b) What would happen if this dataset is loaded into a SQL table with Sales_Amount as DECIMAL?

Answer:

Sales_Amount should be numeric.

Record O104 contains "Three Thousand" which will fail numeric validation.

If loaded into DECIMAL column, the load will fail or throw a conversion error.

Q5. Date Format Consistency

The Order_ID column has multiple formats.

a) List all date formats present in the dataset

b) Why is this a problem during data loading?

Answer:

Date formats present:

1. DD-MM-YYYY (12-01-2024)

2. YYYY/MM/DD (2024/01/18)

Problem: Database expects consistent format; inconsistent formats may cause load errors or incorrect sorting.

Q6. Load Readiness Decision

Based on the dataset condition:

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a) Should this dataset be loaded directly into the database? (Yes/No)

b) Justify your answer with at least three reasons

Answer:

1. No, the dataset should not be loaded directly.
2. Reasons: Duplicate Primary Key, Missing Value, Invalid Data Type, Inconsistent Date Format.

Q7. Pre-Load Validation Checklist

List the exact pre-load validation checks you would perform on this dataset before loading.

Answer:

1. Primary Key uniqueness check.

2. NULL value validation.

3. Data type validation.

4. Date format validation.

5. Duplicate removal.

6. Numeric range validation.

Q8. Cleaning Strategy

Describe the step-by-step cleaning actions required to make this dataset load-ready

Answer:

Step 1: Remove duplicate O101 record.

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Step 2: Handle NULL value in O102 (remove or update).

Step 3: Convert "Three Thousand" to 3000.

Step 4: Standardize all dates to YYYY-MM-DD format.

Step 5: Revalidate and prepare for loading.

Q9. Loading Strategy Selection

Assume this dataset represents daily sales data.

a) Should a Full Load or Incremental Load be used?

b) Justify your choice.

Answer:

1. Incremental Load should be used.
2. Because daily sales data increases regularly and only new records should be loaded.

Q10. BI Impact Scenario

Assume this dataset was loaded without cleaning and connected to a BI dashboard.

a) What incorrect results might appear in Total Sales KPI?

b) Which records specifically would cause misleading insights?

c) Why would BI tools not detect these issues automatically?

Answer:

1. Total Sales KPI may show incorrect results due to duplicates and invalid values.
2. Problematic Records: O101 (duplicate), O102 (NULL), O104 (text value), O103 (date format).

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3. BI tools do not automatically detect data quality issues; they calculate based on stored data.